

In 7

Review

$$1 (x-1)(2x^2+3x-2)$$

$$2 \ln 1-2x = 0.4$$

$$1-2x = e^{0.4}$$
$$x = \frac{1-e^{0.4}}{2}$$

$$3 \begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} 9 \\ k \end{pmatrix} k = -15$$

$$4 a_6 = 33$$

$$\sum_1^6 = 2+3+5+9+17$$

$$= 36$$

$$5 g(x) = x^2 - x - 6$$

$$g(-4) = 16 + 4 - 6$$
$$= 14$$

$$g(x) = 6$$

$$x^2 - x - 6 = 6$$

$$x^2 - x - 12 = 0$$

$$(x+3)(x-4) = 0$$

$$x = -3 \text{ or } x = 4$$

Consolidation

$$1 \int_0^1 \frac{2x+3}{x^2+3x+2} dx \quad (x+2)(x+1)$$

$$= \left[\ln(x^2+3x+2) \right]_0^1$$

$$= \ln(1+3+2) - \ln 2$$

$$= \ln 3$$

$$2 a) \frac{5}{(2x+1)(x+3)} = \frac{2}{2x+1} - \frac{1}{x+3}$$

$$A(x+3) + B(2x+1) = 5$$

$$-5B = 5 \quad B = -1$$

$$A = 2$$

$$6) \left[\ln 2x+1 - \ln x+3 \right]'_0$$

$$\ln 3 - \ln 4 + \ln 3$$

$$\frac{2 \ln 3 - \ln 4}{\ln 4}$$

$$3 a) \frac{3x+5}{(x+1)(x+2)(x+3)} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{x+3}$$

$$A(x+2)(x+3) + B(x+1)(x+3) + C(x+1)(x+2) = 3x+5$$

$$\begin{aligned} A &= 1 \\ B &= 1 \\ C &= -2 \end{aligned} \quad \frac{1}{x+1} + \frac{1}{x+2} - \frac{2}{x+3}$$

$$b) \int \frac{1}{x+1} dx + \int \frac{1}{x+2} dx - \int \frac{2}{x+3} dx$$

$$= \left[\ln|x+1| + \ln|x+2| - 2 \ln|x+3| \right]_0^2$$

$$\ln 3 + \ln 4 - 2 \ln 5 - \ln 2 - \ln 2 + 2 \ln 3$$

$$3 \ln 3 - 2 \ln 2 + \ln 4 - 2 \ln 5$$

$$4 a) \frac{16}{(x+1)^2(x-3)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-3}$$

$$A(x+1)(x-3) + B(x-3) + C(x+1)^2 = 16$$

$$C = 1$$

$$B = -4$$

$$A = -1$$

$$\therefore \frac{16}{(x+1)^2(x-3)} = \frac{-1}{x-3} - \frac{4}{(x+1)^2} - \frac{1}{x+1}$$

$$b) \int_1^2 \frac{16}{(x+1)^2(x-3)} dx$$

$$= \left[\ln|x-3| - \ln|x+1| + 4(x+1)^{-1} \right]_1^2$$

$$= -\ln 1 - \ln 3 + \frac{4}{3} - (\ln 2 - \ln 2 - 2)$$

$$= -\ln 3 - \frac{2}{3}$$

Exam Questions

$$1) f(x) = \frac{x-4}{(x+2)(x-1)} + \frac{3x+1}{(x+3)(x-1)}$$

$$\begin{aligned} a) &= \frac{(x-4)(x+3) + (3x+1)(x+2)}{(x+2)(x-1)(x+3)} \\ &= \frac{x^2 - x - 12 + 3x^2 + 7x + 2}{(x+2)(x-1)(x+3)} \\ &= \frac{4x^2 + 6x - 10}{(x+2)(x-1)(x+3)} \\ &= \frac{(2x+5)(2x-2)}{(x+2)(x-1)(x+3)} \\ &= \frac{2(2x+5)}{(x+2)(x+3)} \end{aligned}$$

$$b) \int_0^{a+4} f(x) dx = 2 \ln 3$$

$$\frac{2(2x+5)}{(x+2)(x+3)} = \frac{A}{x+2} + \frac{B}{x+3}$$

$$A(x+3) + B(x+2) = 2(2x+5)$$

$$A = 2$$

$$-B = -2$$

$$B = 2$$

$$\frac{2}{x+2} + \frac{2}{x+3}$$

$$\int_0^{a+4} \left(\frac{2}{x+2} + \frac{2}{x+3} \right) dx$$

$$= \left[2 \ln|x+2| + 2 \ln|x+3| \right]_0^{a+4}$$

$$2 \ln|a+6| + 2 \ln|a+7| - 2 \ln|a+2| - 2 \ln|a+3| = 2 \ln 3$$

$$2 \left(\ln \left(\frac{(a+6)(a+7)}{(a+2)(a+3)} \right) \right) = 2 \ln 3$$

$$(a+6)(a+7) = 3(a+2)(a+3)$$

$$a^2 + 13a + 42 = 3a^2 + 15a + 18$$

$$2a^2 + 2a - 24 = 0$$

$$a^2 + a - 12 = 0$$

$$(a+4)(a-3) = 0$$

$$\therefore a = -4 \text{ or } a = 3$$

$$\begin{aligned} 6 \leq a < 7 \\ \therefore a = 3 \end{aligned}$$